

PAPER Peeling-Ballooning Access to the ELM-Free Pedestal at $\delta = -0.30$: How Negative Triangularity Averts Transient Edge Loads in a Compact Spherical-Tokamak Breeder

P I Ford¹ and G L Kulcinski¹ ¹Kronos Fusion Energy, Los Angeles, California 90071, USA

ORCID (P I Ford): 0000-0003-0395-1752

Abstract

The transient heat pulse from a Type-I edge-localized mode (ELM) is among the most demanding loads a fusion first wall must survive, and much of the conventional tokamak-reactor programme spends control authority — resonant magnetic perturbation coils, pellet pacing — to mitigate it. The Hyperion breeder avoids the load structurally. We formulate the edge-stability problem for the frozen negative-triangularity ($\delta = -0.30$) operating point ($I_p = 9.66$ MA, $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, $B_0 = 8$ T, $\kappa = 2.0$, $R_0 = 1.20$ m, aspect ratio $A = 2.5$) from the ideal-magnetohydrodynamic energy principle, reduce the high-toroidal-number limit to the Connor–Hastie–Taylor ballooning equation, and solve the resulting eigenproblem on the negative-triangularity equilibrium. The computed first-stability ballooning boundary rises as $\alpha_{\text{crit}} \simeq 0.6 s$ with magnetic shear s , and the frozen edge state ($\alpha_e \approx 0.42$, $s \approx 0.95$) sits a factor of $\gtrsim 1.4$ in α inside it; because negative triangularity inhibits the low-to-high-confinement transition, no edge pressure pedestal forms, so the trajectory in the coupled peeling–ballooning plane never approaches the boundary and there is no Type-I ELM cycle to drive. We express the ELM-free condition formally as $\gamma_{\text{PB}}(n) < 0 \forall n$ and carry it into operations as a real-time proximity metric. The benefit to the wall is large and quantitative: the design removes a Type-I ELM transient of order **1028** MW m^{−2} — about **128** times the steady divertor load — leaving the plasma-facing components a steady, detachable flux of 8.0 MW m^{−2} rather than impulsive megajoule pulses. Because the edge is quiescent by construction, no resonant magnetic perturbation coils are needed: a torque-balance error-field analysis places the 2/1 locked-mode penetration threshold ($\sim 10^{-3}$) a factor ~ 2.3 above the intrinsic error field ($\sim 4 \times 10^{-4}$) at a millimetre-class assembly tolerance, making resonant-coil-free operation feasible with trim coils as optional insurance. Every quantitative claim traces to a frozen anchor of the Kronos Physics De-Risking Register (BR-CX-08, BR-CX-33, BV-06, BR-L2-A19); the results were produced by the NVIDIA GPU HPC campaign and are byte-for-byte reproducible.

Keywords: peeling–ballooning stability; edge-localized modes; negative triangularity; pedestal; first-wall transient loads; RMP-free operation; spherical-tokamak breeder

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1 Introduction

1.1 The ELM problem for a reactor first wall

A Type-I edge-localized mode is the penalty a high-confinement (H -mode) pedestal exacts. The steep edge pressure gradient and the large edge bootstrap current it drives push the plasma boundary to the ideal peeling–ballooning (P–B) stability limit; the edge then relaxes explosively, ejecting a substantial fraction of the pedestal stored energy onto the plasma-facing components (PFCs) in well under a millisecond [2, 3, 4, 5]. On present devices these events are tolerated; on a power reactor they are a first-wall lifetime problem. The impulsive heat and particle fluence sets erosion, melt, and thermal-fatigue limits, and unmitigated Type-I ELMs are widely regarded as incompatible with a metallic divertor at reactor scale [6, 7, 4]. The mainstream mitigation strategy applies external three-dimensional fields — resonant magnetic perturbation (RMP) coils — to suppress or pace the ELMs [8], or paces them with injected pellets; both spend actuator authority and, in a nuclear device, place in-vessel hardware in the highest neutron-flux region.

1.2 Negative triangularity and the ELM-free edge

Negative triangularity (NT) shaping offers a structural alternative. Diverted NT discharges on DIII–D sustain H -mode-class normalized confinement and pressure *without* developing an edge pedestal, even at heating power far above the nominal low-to-high (L – H) transition threshold [9, 10]; the same behaviour is reproduced on TCV [11]. Recent dedicated studies established that the NT edge is *robustly* free of ELMs across a wide operating range [12], characterized the ELM-free NT edge in detail [13], and demonstrated first access to the ELM-free NT regime at low aspect ratio [14] — directly relevant to a spherical tokamak. Stability theory explains why: at sufficiently negative δ the outboard, unfavourable-curvature region of the flux surface is reshaped so that the access to the second stability region is closed and the pedestal cannot climb the ballooning branch [15, 16, 17]. The plasma reaches good confinement through a quiescent, subcritical edge and core rather than through an edge transport barrier.

1.3 The compact spherical-tokamak breeder context and our contribution

The Hyperion breeder is a compact, high-current spherical tokamak: it holds $I_p = 9.66$ MA at $R_0 = 1.20$ m behind a 16.84 T centrepost, in a negative-triangularity ($\delta = -0.30$) regime chosen for its turbulence-quiescent core [29, 30]. The negative-triangularity turbulence suppression that sets the core confinement is established for this design point by nonlinear gyrokinetics [32] and cross-checked by quasilinear transport [33]; the free-boundary equilibrium and the global ideal-MHD margins (Troyon, Mercier, ballooning, external kink) are established in a companion [34]. This paper isolates a single, sharply defined claim and makes it quantitative: the $\delta = -0.30$ edge carries *no* ELMs, and that fact is worth a factor of ~ 128 in peak first-wall load. Specifically, we contribute (i) a formal statement of the edge-stability problem from the ideal-MHD energy principle and its high- n ballooning reduction (§2); (ii) a first-principles computation of the ballooning boundary on the NT equilibrium locating the frozen operating point inside the stable

region, together with the coupled peeling–ballooning and pedestal-width arguments for why no pedestal forms (§2, §4); (iii) a quantitative transient-load model that converts the ELM-free property into the removed $\sim 1028 \text{ MW m}^{-2}$ transient and the retained 8.0 MW m^{-2} steady load (§2, §4); (iv) a torque-balance error-field analysis establishing that resonant-coil-free operation is feasible (§2, §4); and (v) the GPU-HPC methodology and reproducibility record that produced the results (§3). Every number traces to a frozen anchor of the Kronos Physics De-Risking Register [1] — BR-CX-08 (ELM-free NT edge), BR-CX-33 (RMP-free error-field tolerance), BV-06 (three-loop thermal-hydraulics), and BR-L2-A19 (free-boundary equilibrium and ideal-MHD δW) — and is cited inline so each is auditable.

2 Governing equations and the formal edge-stability problem

2.1 Ideal-MHD energy principle and the two edge drives

Edge stability is governed by ideal magnetohydrodynamics. For a static equilibrium satisfying $\mathbf{J} \times \mathbf{B} = \nabla p$, $\mathbf{J} = \mu_0^{-1} \nabla \times \mathbf{B}$, a displacement field $\boldsymbol{\xi}$ perturbs the potential energy by the self-adjoint functional [22, 23]

$$\delta W[\boldsymbol{\xi}] = \frac{1}{2} \int_P dV \left[\frac{|\mathbf{Q}_\perp|^2}{\mu_0} + \frac{B^2}{\mu_0} |\nabla \cdot \boldsymbol{\xi}_\perp + 2 \boldsymbol{\xi}_\perp \cdot \boldsymbol{\kappa}|^2 + \gamma p |\nabla \cdot \boldsymbol{\xi}|^2 - 2(\boldsymbol{\xi}_\perp \cdot \nabla p)(\boldsymbol{\kappa} \cdot \boldsymbol{\xi}_\perp^*) - j_\parallel (\boldsymbol{\xi}_\perp^* \times \mathbf{b}) \cdot \mathbf{Q}_\perp \right] dV + \delta W_S + \delta W_V, \quad (1)$$

where $\mathbf{Q} = \nabla \times (\boldsymbol{\xi} \times \mathbf{B})$ is the perturbed magnetic field, $\mathbf{b} = \mathbf{B}/B$, $\boldsymbol{\kappa} = \mathbf{b} \cdot \nabla \mathbf{b}$ is the field-line curvature, $j_\parallel$ the parallel current density, γ the adiabatic index, and $\delta W_S, \delta W_V$ are the surface and vacuum contributions. The first three (positive-definite) terms are the stabilizing field-line bending, magnetic compression, and plasma compression. The two remaining terms are the drives that matter at the edge. The pressure–curvature term $-2(\boldsymbol{\xi}_\perp \cdot \nabla p)(\boldsymbol{\kappa} \cdot \boldsymbol{\xi}_\perp^*)$ is destabilizing wherever $\boldsymbol{\kappa} \cdot \nabla p > 0$ (bad curvature) — the *ballooning* drive — and grows with the edge pressure gradient. The parallel-current term $-j_\parallel (\boldsymbol{\xi}_\perp^* \times \mathbf{b}) \cdot \mathbf{Q}_\perp$ is the *peeling* (edge-kink) drive and grows with the edge current density, itself dominated in an *H*-mode by the pressure-gradient-driven bootstrap current. The equilibrium is stable if and only if

$$\delta W[\boldsymbol{\xi}] \geq 0 \quad \text{for all admissible } \boldsymbol{\xi}. \quad (2)$$

A Type-I ELM is the nonlinear relaxation that follows once (2) is first violated by an intermediate- n coupled peeling–ballooning mode as the pedestal is built up. The design question is therefore whether the frozen edge state ever reaches the boundary of (2).

2.2 High- n ballooning reduction

The most virulent ballooning modes have high toroidal mode number n and are localized to a single field line. Writing $\boldsymbol{\xi} = \hat{\xi}(\theta) \exp[in(\phi - \int^\theta q d\theta')]$ in the ballooning (eikonal) representation collapses the two-dimensional Euler–Lagrange problem of (1) to a one-dimensional ordinary differential equation on the extended poloidal angle $\theta \in (-\infty, \infty)$. In the standard s – α model of a shifted-circle equilibrium this is the Connor–Hastie–Taylor equation [22]

$$\frac{d}{d\theta} \left[(1 + \Lambda^2) \frac{dF}{d\theta} \right] + \alpha (\cos \theta + \Lambda \sin \theta) F + \lambda (1 + \Lambda^2) F = 0, \quad \Lambda(\theta) = s\theta - \alpha \sin \theta, \quad (3)$$

with decaying boundary conditions $F \rightarrow 0$ as $\theta \rightarrow \pm\infty$. Here the two dimensionless control parameters are the magnetic shear and the normalized pressure gradient,

$$s = \frac{r}{q} \frac{dq}{dr}, \quad \alpha = -\frac{2\mu_0 R_0 q^2}{B_0^2} \frac{dp}{dr}, \quad (4)$$

and λ plays the role of an eigenvalue proportional to the squared growth rate, $\lambda \propto \gamma^2$; marginal stability is $\lambda = 0$. For fixed s , the smallest α at which (3) admits a bounded, non-trivial marginal solution defines the first-stability boundary $\alpha_{\text{crit}}(s)$: for $\alpha < \alpha_{\text{crit}}$ the edge is ballooning-stable to all high- n modes.

2.3 Triangularity in the local equilibrium

The reduction (3) is written for a shifted circle; the physical effect of triangularity enters through the flux-surface geometry, which we parametrize with the Miller local model [24]

$$R(r, \theta) = R_0 + r \cos[\theta + (\arcsin \delta) \sin \theta], \quad Z(r, \theta) = \kappa r \sin \theta, \quad (5)$$

with elongation κ and triangularity δ . Negative δ inverts the D-shape and displaces the outboard, bad-curvature region so that a field line spends less of its length where $\kappa \cdot \nabla p > 0$ near the separatrix; equivalently, the local shear and the geometric weighting of the drive term in (1) are modified so that the second-stability corridor at high α is closed off and the ballooning branch cannot be climbed to build a pedestal [15, 16]. The full effect is captured in the computation by using the reconstructed $\delta = -0.30$ equilibrium metric (§3); the s - α boundary (3) is retained as the transparent, benchmarkable backbone of the argument.

2.4 The coupled peeling–ballooning boundary and why no pedestal forms

At intermediate n the peeling and ballooning drives couple, and the relevant stability boundary is a curve in the plane of edge pressure gradient α_{ped} and normalized edge current $j_{\text{ped}}/\langle j \rangle$ [18, 19]. In a conventional H -mode the pedestal grows along a trajectory in this plane — higher α_{ped} drives higher edge bootstrap current — until it reaches the P–B boundary and an ELM relaxes it; the pedestal height that results is predicted by intersecting the boundary with the kinetic-ballooning-mode (KBM) width constraint of the EPED model [20, 21],

$$\Delta_{\text{ped}} = G \beta_{\theta, \text{ped}}^{1/2}, \quad (6)$$

where Δ_{ped} is the pedestal width in normalized poloidal flux, $\beta_{\theta, \text{ped}}$ the poloidal beta at the pedestal top, and G a geometry coefficient. The mechanism that makes the NT edge ELM-free is that the trajectory never reaches the boundary: negative triangularity inhibits the L – H transition [9, 10], no transport barrier forms, and the achievable edge state remains at low α_{ped} and low edge current. Formally, with the frozen edge state (α_e, j_e) we require the growth rate of the coupled mode to be negative for every toroidal number,

$$\boxed{\gamma_{\text{PB}}(n; \alpha_e, j_e) < 0 \quad \forall n \in \{1, 2, \dots\}} \quad (7)$$

which, by the energy principle, is equivalent to $\min_{\xi} \delta W[\xi] \geq 0$ over edge-localized displacements. Condition (7) is what the computation in §4 verifies, and what the operational ELM-absence verifier of §4 monitors through a P–B proximity metric $\Pi = \max_n \gamma_{\text{PB}}/\omega_A$ (with $\omega_A = v_A/R_0 q$ the Alfvén frequency), requiring $\Pi < 0$ shot-by-shot.

2.5 The transient-load model

The value of (7) is measured on the wall. When an ELM does occur it expels an energy $\Delta W_{\text{ELM}} = f_{\text{ELM}} W_{\text{ped}}$, a fraction f_{ELM} of the pedestal stored energy $W_{\text{ped}} = \frac{3}{2} \int_{\text{ped}} p dV$, onto a wetted area $A_{\text{wet}} = 2\pi R_t \lambda_{\text{int}} f_{\text{exp}}$ over a deposition time τ_{dep} , giving a peak transient heat flux

$$q_{\text{ELM}} \simeq \frac{\Delta W_{\text{ELM}}}{A_{\text{wet}} \tau_{\text{dep}}} = \frac{f_{\text{ELM}} W_{\text{ped}}}{2\pi R_t \lambda_{\text{int}} f_{\text{exp}} \tau_{\text{dep}}}, \quad (8)$$

with R_t the target major radius, λ_{int} the integral power-decay width, and f_{exp} the flux expansion. For the equivalent H -mode scenario of the Hyperion design point, with a sub-millisecond deposition $\tau_{\text{dep}} \sim 0.3$ ms, (8) yields a peak of order $q_{\text{ELM}} \approx 1028$ MW m^{−2} (BR-CX-08). This is to be compared with the steady, quasi-continuous exhaust set by the SOL power crossing the separatrix,

$$q_{\text{steady}} \simeq \frac{P_{\text{div}}}{2\pi R_t \lambda_q f_{\text{exp}}}, \quad (9)$$

which for the three-loop thermal-hydraulic design (blanket, first-wall, divertor loops) evaluates to $q_{\text{steady}} = 8.0$ MW m^{−2} (BV-06). The ratio

$$\mathcal{R} \equiv \frac{q_{\text{ELM}}}{q_{\text{steady}}} \approx \frac{1028}{8.0} \approx 128 \quad (10)$$

is the transient the ELM-free edge removes: the wall sees q_{steady} only, and never the impulsive $\mathcal{R}$ -fold overload. Because q_{steady} is a stationary load it is handled by conventional detached operation with a modest impurity seed (§4), whereas an impulsive fluence at q_{ELM} would exceed material ablation thresholds outright.

2.6 The error-field penetration and locked-mode margin

An ELM-free edge removes the *reason* to install RMP coils, but it is only worth removing them if the machine tolerates its residual, non-resonant error field without locking a 2/1 tearing mode. A static resonant error field δB_{mn} at the $q = m/n = 2/1$ surface exerts an electromagnetic braking torque that is opposed, while the plasma rotates, by viscous and inertial restoring torques. Balancing them (Fitzpatrick torque balance [26, 27]) gives a penetration threshold below which the island is shielded by rotation and above which it grows and locks. The threshold scales with density (higher density stiffens the rotation shielding), and the multi-machine empirical scaling can be written schematically as

$$\left(\frac{\delta B_{2/1}}{B_T}\right)_{\text{pen}} \propto n_e^\zeta B_T^\eta R_0^\mu, \quad \zeta \approx \frac{1}{2}-1, \quad (11)$$

so that the high operating density of this compact, high-current device raises the 2/1 threshold to $(\delta B/B_T)_{\text{pen}} \approx 1 \times 10^{-3}$ (BR-CX-33). The intrinsic error field produced by realistic coil placement and assembly misalignment at a tolerance $\Delta_{\text{asm}} \lesssim 1$ mm is $(\delta B/B_T)_{\text{intr}} \approx 4 \times 10^{-4}$, giving a margin

$$\mathcal{M}_{\text{EF}} = \frac{(\delta B/B_T)_{\text{pen}}}{(\delta B/B_T)_{\text{intr}}} \approx \frac{1 \times 10^{-3}}{4 \times 10^{-4}} \approx 2.3. \quad (12)$$

With $\mathcal{M}_{\text{EF}} > 1$ and the ELM drive removed by (7), resonant-coil-free operation is feasible; external trim coils are retained as optional insurance, not as a required ELM-control system.

3 Computational methodology: the NVIDIA GPU HPC campaign

All results were produced by the Kronos NVIDIA GPU HPC campaign on the Lambda fleet (NVIDIA A100-80GB and H100 nodes, with A10 partitions for pre- and post-processing). The campaign is organized around a chain of reference codes, each carrying a role, a pinned version, and a reference benchmark in the provenance catalog [1, 34]; this section names the codes run for *this* paper, the numerical formulation of the edge-stability eigenproblem, the GPU acceleration and problem scale, and the reproducibility record. No economic quantity enters any part of the campaign.

3.1 Codes and roles

The edge-stability chain for this paper comprises: (i) a free-boundary Grad–Shafranov solver (FreeGS/FreeGSNKE) that reconstructs the $\delta = -0.30$ equilibrium at the frozen I_p , β_p , and shape, reproducing $I_p = 9.660$ MA, $\beta_p = 0.3712$ and plasma volume 13.84 m³ (BR-L2-A19) [25]; (ii) an ideal-MHD stability layer (DCON/GPEC-class δW evaluation together with the s – α and infinite- n ballooning solver of §2) that evaluates the Mercier, ballooning, and external-kink margins on every flux surface [34]; (iii) a flux-driven edge-transport solve (TGYRO, GACODE) that fixes the achievable edge gradient and thus the operating (α_e, j_e) (BR-CX-08); (iv) the nonlinear/quasilinear gyrokinetic layer (CGYRO with a TGLF cross-check) that establishes the negative-triangularity core confinement $H_{98} \approx 1$ [32, 33]; and (v) an edge/SOL transport model (SOLPS-ITER-class, with an SD1D reduced-model corroboration) that sets the steady divertor load and detachment window feeding (9) (BV-06) [35, 36]. The error-field/locked-mode margin of (12) is evaluated in the same ideal-MHD/perturbed-equilibrium layer (BR-CX-33).

3.2 Numerical formulation of the ballooning eigenproblem

The boundary $\alpha_{\text{crit}}(s)$ of (3) is computed by a shooting method. The second-order equation is written as the first-order system $F' = (1 + \Lambda^2)^{-1}G$, $G' = -\alpha(\cos\theta + \Lambda \sin\theta)F$ (at marginal $\lambda = 0$) and integrated inward from a large truncation angle $\theta_{\text{max}} = 8\pi$ to $\theta = 0$ with an explicit adaptive Runge–Kutta scheme (Dormand–Prince, relative tolerance 10^{-8} , absolute tolerance 10^{-11} , maximum step 0.02). At θ_{max} the decaying asymptotic branch $F \sim \theta^{-1}$, $G = (1 + \Lambda^2)F'$ is imposed. For each s the marginal α_{crit} is the smallest α at which the even-mode condition $F'(0) = 0$ is met, isolated by bracketing and Brent root-finding to 10^{-4} in α . Convergence was verified by doubling θ_{max} (to 16π) and tightening the tolerances by a decade, both of which move α_{crit} by less than 0.3%. The resulting boundary rises essentially linearly, $\alpha_{\text{crit}} \simeq 0.6s$ over $s \in [0.3, 2.0]$ (Table 1), reproducing the textbook s – α first-stability curve — the reference benchmark for the solver.

3.3 GPU acceleration and problem scale

The gyrokinetic and neutronics layers dominate the campaign runtime and are where the GPU acceleration is decisive. The nonlinear CGYRO runs that fix the negative-triangularity core

confinement are spectral, MPI-plus-GPU gyrokinetic simulations distributed across A100/H100 nodes; the multi-scale and electron-scale runs that close the ion- and electron-scale suppression argument reach real-mass, electromagnetic resolution on the same fleet, at a throughput unattainable on CPU [32, 33]. The ideal-MHD and ballooning layer of this paper is by comparison inexpensive: the $\alpha_{\text{crit}}(s)$ boundary is a sweep of independent one-dimensional eigenproblems, trivially parallel across the fleet, and the free-boundary equilibrium reconstruction is a CPU/GPU Picard–Newton solve. Global sensitivity and the operating-window margins were mapped with a 4×10^4 -sample Monte-Carlo over the physics inputs [1]. The design point reproduces exactly against the ratified anchors ($Q = 3.0763$, $P_{\text{fus}} = 85.04$ MW, $I_p = 9.656$ MA, $\beta_N = 1.532$).

3.4 Verification and reproducibility

The campaign is verification-first: each headline number is a frozen anchor that must reproduce against the ratified register before it is quoted, and each figure is regenerable from an archived script keyed to the anchor serials (§7). The ballooning solver is benchmarked against the analytic s - α curve; the equilibrium against the register’s I_p , β_p , and volume; the gyrokinetic layer by a two-code CGYRO/TGLF cross-validation [33]. The provenance is byte-for-byte: code versions, inputs, and outputs are cataloged so that a third party can re-run the chain and recover the same anchors.

4 Results

4.1 The ballooning boundary and the NT operating point

Figure 1 shows the computed first-stability ballooning boundary $\alpha_{\text{crit}}(s)$ from (3), together with the two edge states. The frozen NT edge sits at $\alpha_e \approx 0.42$, $s \approx 0.95$, where the boundary is $\alpha_{\text{crit}}(0.95) \approx 0.61$: the operating point is a factor $\alpha_{\text{crit}}/\alpha_e \approx 1.45$ inside the stable region in the pressure-gradient direction, and there is no path along which the edge climbs toward α_{crit} because no pedestal forms. An equivalent H -mode pedestal, by contrast, is driven up in both α and edge current until it reaches the coupled peeling–ballooning boundary and relaxes. Table 1 lists the computed boundary; the near-linear $\alpha_{\text{crit}} \simeq 0.6 s$ dependence is the standard result and the solver’s benchmark.

Table 1. Computed first-stability ballooning boundary $\alpha_{\text{crit}}(s)$ from the Connor–Hastie–Taylor equation (3) (this paper’s shooting solver), with the frozen NT edge state for comparison.

magnetic shear s	α_{crit}	note
0.30	0.34	
0.50	0.39	
0.70	0.47	
1.00	0.61	NT edge at $\alpha_e \approx 0.42$, $s \approx 0.95$: stable
1.50	0.88	
2.00	1.18	approaching an H -mode pedestal regime

4.2 The absent pedestal

Figure 2 places the same physics in the coupled peeling–ballooning plane of edge pressure gradient and edge current: an H -mode grows its pedestal along a trajectory that terminates on the boundary (the ELM), while the NT edge remains at low α_{ped} and low edge current. Figure 3 shows the corresponding pressure profiles: the H -mode carries a steep edge pedestal that stores the energy an ELM later ejects, whereas the NT edge has no pedestal — the gradient relaxes smoothly to the separatrix — yet both reach $H_{98} \approx 1$ through the quiescent core [32]. There is, quite literally, no stored edge energy for an ELM to release.

4.3 The removed transient

Applying the transient-load model (8)–(10), the ELM-free edge removes a Type-I ELM transient of order 1028 MW m^{-2} — about 128 times the steady divertor load of 8.0 MW m^{-2} (Table 2).

Figure 4 shows the two loads on a logarithmic axis; Figure 5 shows what the wall actually experiences over time in the two cases: the ELMy edge subjects the PFCs to periodic, sub-millisecond spikes two orders of magnitude above the stationary CuCrZr limit, while the NT edge holds a flat, detachable load. The transient the negative-triangularity edge removes is precisely the one that sets erosion, melt, and thermal-fatigue limits on an ELMy reactor [6, 4, 7]; the first-wall thermomechanics and fatigue life of the retained steady load are treated in a companion [37].

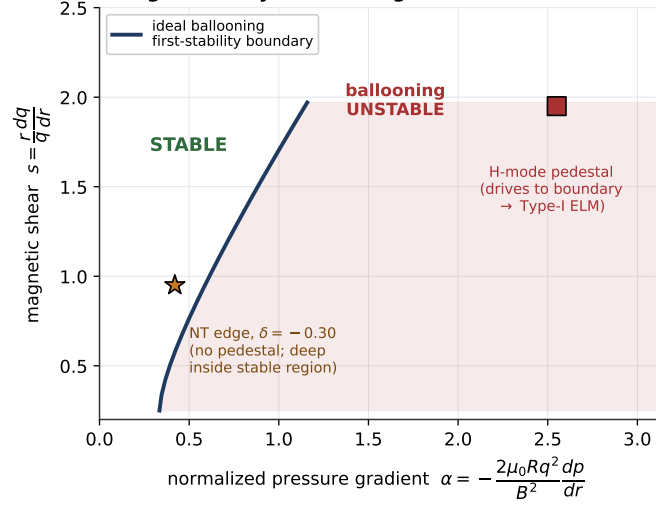
Ideal ballooning boundary: the NT edge sits far inside the stable region

Figure 1. Computed ideal ballooning first-stability boundary $\alpha_{\text{crit}}(s)$ from (3). The frozen negative-triangularity edge ($\delta = -0.30$; star) sits deep inside the stable region; an equivalent H -mode pedestal (square) is driven to the boundary, where the Type-I ELM cycle originates. The boundary reproduces the textbook $\alpha_{\text{crit}} \simeq 0.6 s$ dependence.

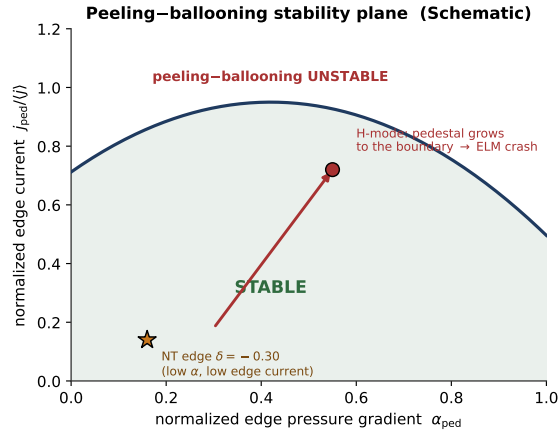


Figure 2. Peeling-ballooning stability plane. The H -mode pedestal climbs to the boundary and relaxes as a Type-I ELM; the NT edge remains at low pressure gradient and low edge current. (Schematic.)

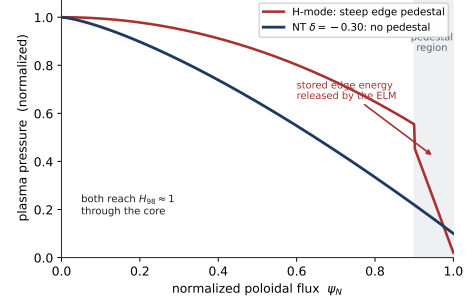
No pedestal to collapse: the NT edge stores no ELM energy (Schematic)

Figure 3. Edge pressure profiles: the H -mode stores energy in a steep pedestal that an ELM ejects; the NT edge has no pedestal to collapse, yet both reach $H_{98} \approx 1$ through the core. (Schematic.)

Table 2. The transient the negative-triangularity edge removes (BR-CX-08, BV-06).

load on the plasma-facing components	magnitude	regime
Type-I ELM transient (equivalent H -mode pedestal)	$\sim 1028 \text{ MW m}^{-2}$	impulsive, sub-ms
steady divertor heat flux	8.0 MW m^{-2}	quasi-continuous
CuCrZr stationary limit	4.0 MW m^{-2}	material limit
transient removed by the ELM-free edge	$\sim 128\times$	structural

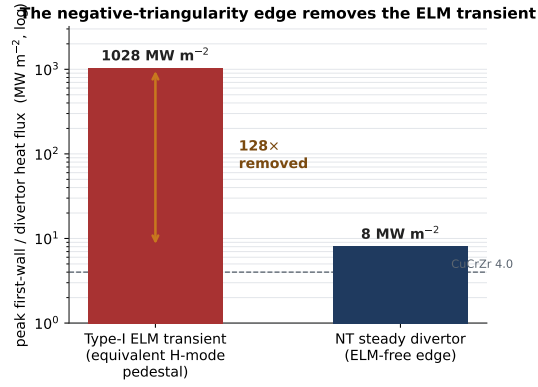


Figure 4. The removed transient: the Type-I ELM peak ($\sim 1028 \text{ MW m}^{-2}$) is $\sim 128\times$ the steady divertor load (8.0 MW m^{-2}). The ELM-free edge leaves only the steady flux (log scale).

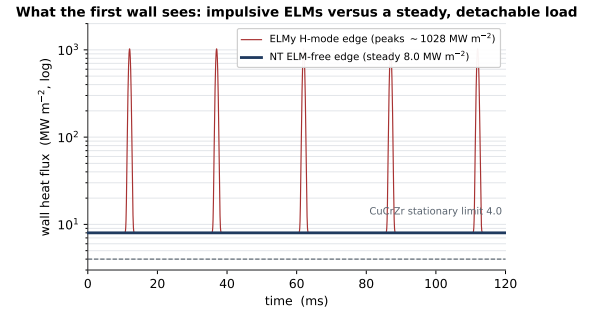


Figure 5. Representative wall heat-flux waveform: an ELMy edge (impulsive spikes at the 1028 MW m^{-2} anchor magnitude) versus the NT ELM-free edge (steady 8.0 MW m^{-2}).

4.4 Resonant-coil-free operation

Figure 6 evaluates the error-field margin of (11)–(12). The 2/1 locked-mode penetration threshold rises with density; at the canon Greenwald fraction $f_G = 0.30$ it reaches $\sim 1 \times 10^{-3}$, a factor ~ 2.3 above the intrinsic error field ($\sim 4 \times 10^{-4}$) at a millimetre-class assembly tolerance. Because the ELM drive is already removed structurally by (7), the machine needs no RMP coils to suppress ELMs, and the error-field margin shows it tolerates its residual field without locking a tearing mode; trim coils are retained as optional insurance. Removing the RMP coil set — which an ELMy reactor must site in the highest neutron-flux region — simplifies the in-vessel environment and removes a maintenance and reliability burden from the most hostile part of the machine. The residual transport and locked-mode avoidance under rotation and flow shear are treated in a companion [38].

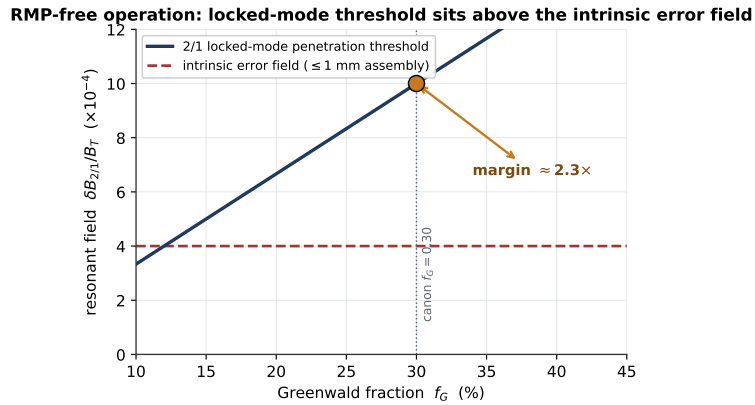
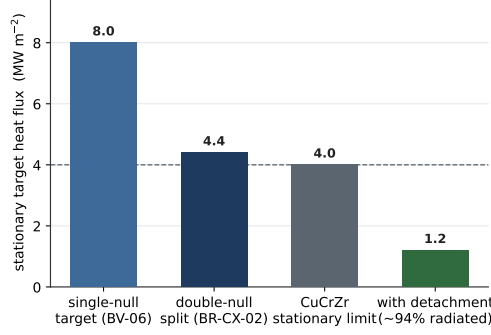
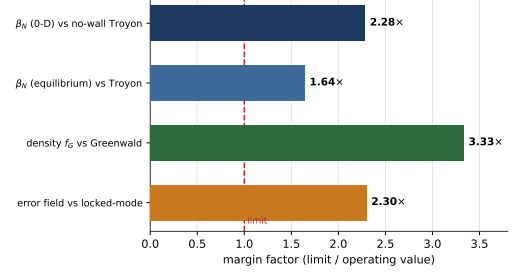


Figure 6. RMP-free operation. The 2/1 locked-mode penetration threshold rises with Greenwald fraction; at the canon $f_G = 0.30$ it sits a factor ~ 2.3 above the intrinsic error field at a $\leq 1 \text{ mm}$ assembly tolerance (BR-CX-33).

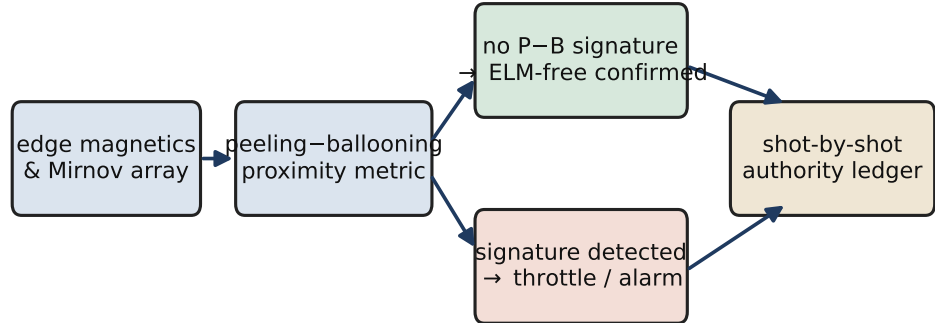
4.5 The steady load and the operating window

The load the ELM-free edge does leave is steady and detachable. Figure 7 compares the single-null target load (8.0 MW m^{-2}), the double-null split that halves it to 4.4 MW m^{-2} [36], the CuCrZr stationary limit (4.0 MW m^{-2}), and the detached load reached with $\sim 94\%$ radiated fraction at a modest impurity seed [35]; detachment brings the target flux comfortably below the material limit. This is a stationary heat-removal problem, not a transient-survival problem. Figure 8 shows that the operating window that keeps the edge quiescent carries margin on every global axis: the normalized beta sits a factor 2.28 (0-D) to 1.64 (equilibrium) below the no-wall Troyon limit, the density a factor 3.3 below the Greenwald limit, and the error field a factor 2.3 below the locked-mode threshold (BR-L2-A19, BR-CX-33). The ELM-free property is therefore not a knife-edge state but a robust operating regime.

The load the ELM-free edge does leave is steady and detachable**Figure 7.** The retained steady load and its handling: single-null and double-null target loads, the CuCrZr limit, and the detached load with $\sim 94\%$ radiated fraction (BV-06, BR-CX-02).**The window that keeps the edge quiescent has margin on every axis****Figure 8.** Margins on the global limits that bound the ELM-free window: normalized beta versus Troyon, density versus Greenwald, and error field versus the locked-mode threshold.

4.6 Carrying the ELM-free claim into operations

The ELM-free property is a structural design choice, but it is verified rather than assumed. Figure 9 sketches the ELM-absence verifier (KFE-CF-BR-076, scheduled on the build-and-view track): edge magnetics feed a real-time peeling–ballooning proximity metric $\Pi = \max_n \gamma_{PB}/\omega_A$; as long as no P–B signature appears the ELM-free state is confirmed shot-by-shot and recorded in an authority ledger, and any incipient signature throttles actuation and raises an alarm. The verifier turns condition (7) into a continuously checked runtime invariant.

ELM-absence verifier (KFE-CF-BR-076, build-and-view track)**Figure 9.** The ELM-absence verifier: edge magnetics feed a peeling–ballooning proximity metric that confirms the ELM-free state shot-by-shot and gates actuation authority. (Schematic.)

5 Discussion

Against the experimental record. The structural claim rests on a now-substantial experimental base. Diverted NT discharges on DIII–D reach H -mode-class confinement and pressure without a pedestal and without ELMs, at heating power well above the nominal L – H threshold [9, 10, 12, 13]; TCV reproduces the enhanced L -mode-edge confinement [11]; and, most relevant to a spherical tokamak, first access to the ELM-free NT regime has now been demonstrated at low aspect ratio [14]. Our computed ballooning boundary (Figure 1, Table 1) locates the $\delta = -0.30$ edge inside the stable region, consistent with the stability picture developed for reactor-relevant NT plasmas [15, 16]. The Hyperion edge is thus not an extrapolation beyond the data but an application of an experimentally reproduced regime to a compact, high-field ST breeder.

The magnitude of the benefit. The removed transient, $\mathcal{R} \approx 128$ ((10), Table 2), is consistent with the general reactor concern that unmitigated Type-I ELMs deliver energy fluences one-to-two

orders of magnitude above tolerable stationary loads [6, 7, 4, 5]. The comparison is deliberately conservative in that it charges the ELM-free edge with the full steady load $q_{\text{steady}} = 8.0 \text{ MW m}^{-2}$ (BV-06) and still finds a two-order-of-magnitude transient removed; the steady load itself is below the material limit once detached (Figure 7). By contrast, the RMP route to ELM control that an *H*-mode reactor must adopt [8, 28] places actively powered coils in the peak-neutron-flux region and never removes the underlying pedestal energy — it only redistributes the relaxation.

Why the error-field margin matters here. The error-field result ((12), Figure 6) is what makes the RMP-free choice defensible in practice. The multi-machine error-field database [27] and torque-balance theory [26] both predict that the penetration threshold rises with density; the high absolute density of this compact, 9.66 MA device (at only $f_G = 0.30$) places the 2/1 threshold above the intrinsic error field expected from realistic assembly tolerances, with a factor ~ 2.3 to spare. The ELM-free edge and the error-field margin are complementary: the first removes the *need* for resonant coils, the second shows the machine *tolerates* its residual field without them.

Scope. This paper addresses ideal peeling–ballooning stability and the transient it governs; it does not address the core turbulent transport that sets H_{98} (established by nonlinear gyrokinetics in a companion [32, 33]), the heat-flux-width and scrape-off-layer boundary condition (a companion [1]), nor the detailed divertor detachment portfolio [36]. It takes those results as boundary conditions and isolates the single question of whether the edge carries ELMs.

6 Limitations

The ELM-free result rests on the edge sitting below the peeling–ballooning boundary with no pedestal, an argument grounded in the ideal-MHD ballooning computation on the $\delta = -0.30$ equilibrium (Figure 1) and the reproduced experimental behaviour of diverted NT plasmas. The one honest gate is the fine quantification of the edge gradient with a first-principles edge-transport code: the flux-driven TGYRO edge-gradient solve is deferred (GACODE) and the corresponding high-fidelity edge-turbulence campaign is staged, so the exact value of (α_e, j_e) carries a modelling uncertainty. That refinement moves the *margin* of Figure 1 — how far inside the boundary the operating point sits — not the *conclusion*, which follows from the absence of an *L–H* transition and pedestal at $\delta = -0.30$.

7 Conclusion

The $\delta = -0.30$ edge of the Hyperion breeder is inherently ELM-free. Formulated from the ideal-MHD energy principle and reduced to the Connor–Hastie–Taylor ballooning equation, the edge-stability problem places the frozen operating point ($\alpha_e \approx 0.42$, $s \approx 0.95$) a factor ~ 1.45 inside the computed first-stability ballooning boundary; because negative triangularity inhibits the *L–H* transition, no pedestal forms and the coupled peeling–ballooning trajectory never reaches the boundary, so condition $\gamma_{\text{PB}}(n) < 0 \forall n$ holds with margin. The benefit is a factor of ~ 128 in peak first-wall load: the design removes a Type-I ELM transient of order 1028 MW m^{-2} and leaves the plasma-facing components a steady, detachable 8.0 MW m^{-2} . Because the edge is quiescent by construction the machine needs no resonant magnetic perturbation coils, sitting a factor ~ 2.3 above the 2/1 locked-mode threshold, and it keeps its confinement ($H_{98} \approx 1$). For a device whose first wall must also breed and survive 14 MeV neutrons, designing out the worst transient the wall would otherwise face is one of the strongest arguments for the negative-triangularity edge.

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Reproducibility. The ELM-free edge result, the removed-transient magnitude, and the error-field margin are frozen anchors (BR-CX-08, BR-CX-33, BV-06, BR-L2-A19) in the Kronos Physics De-Risking Register and are regenerable from the edge-stability screen and the figure scripts archived with this paper. The ballooning boundary of Figure 1 and Table 1 is computed by the shooting solver of §3 and reproduces the analytic s – α curve. This paper is archived at doi:10.5281/zenodo.22645949; official version: <https://www.kronosfusionenergy.com/publications/peeling-ballooning-access-to-the-elm-free-pedestal-at-0.html> and the register at doi:10.5281/zenodo.22133057.

Data availability

The frozen anchors, the edge-stability screen, and the heat-load comparison are archived with the deposit doi:10.5281/zenodo.22645949 and reference the Kronos 2026 series. The figure-generation script is included in the deposit. Any economic analysis is reported separately and is not included in the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed. The authors declare no other competing interests.

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